

Motivation

Reward design is often complex or inaccessible in practice. Imitation Learning overcomes this by learning directly from expert demonstrations, inferring desirable behavior without any environment reward signal. BC, IQ-Learn, CSIL, and CSIL+SOAR are compared on two classic-control tasks, with emphasis on how performance scales with the number of expert trajectories.

Environments

Experiments are conducted on two standard OpenAI Gym classic-control tasks.

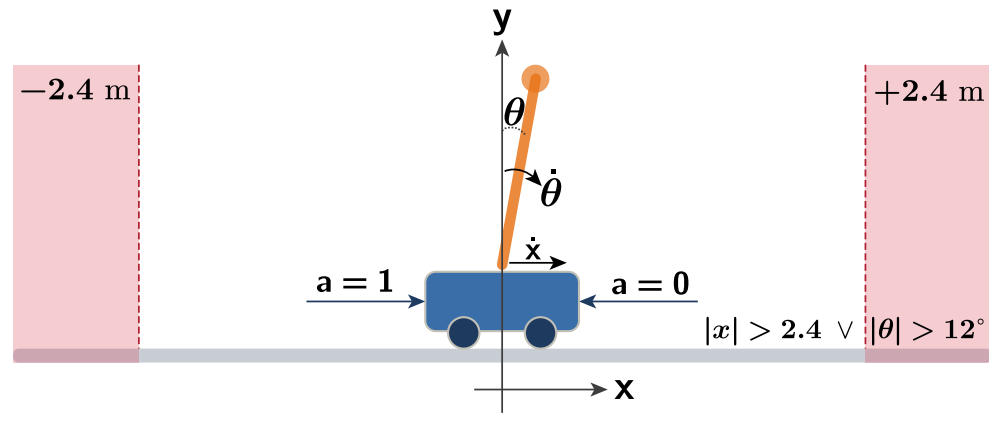


Figure 1. **CartPole-v1**. Balance a pole on a moving cart via discrete left/right force $a \in \{0, 1\}$. State $s \in \mathbb{R}^4$, reward $r_t = +1$ per timestep, truncated at $T = 500$.

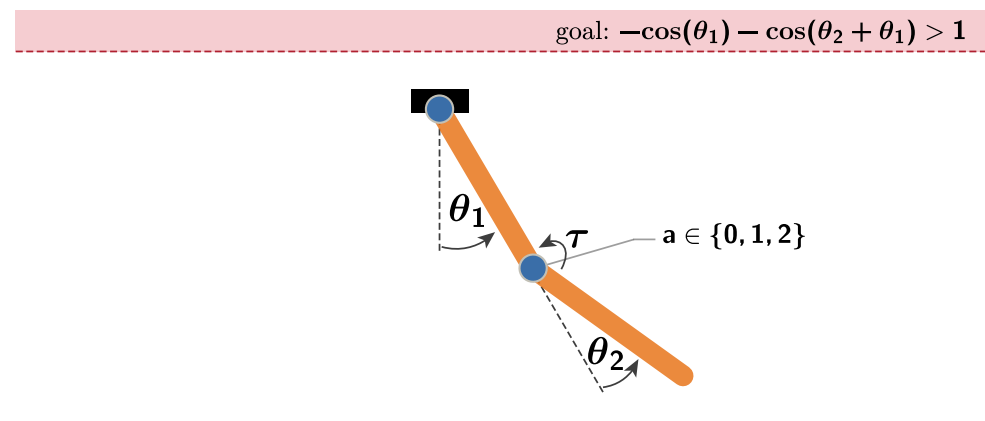


Figure 2. **Acrobot-v1**. Swing the free end of a two-link pendulum above the target height by applying discrete torque $a \in \{0, 1, 2\}$ on the actuated joint. State $s \in \mathbb{R}^6$, reward $r_t = -1$ per timestep until goal is reached.

Methods

Expert trajectories

Proximal Policy Optimization (PPO) [5] requires access to the environment reward and is used solely to generate expert demonstrations. By constraining each update to stay close to the previous policy, it achieves stable training with standard gradient-based optimization. Demonstrations are collected through deterministic rollouts of the fully converged policy.

Raw expert rollouts contain highly correlated consecutive transitions. To reduce this correlation and simulate sparse, realistic expert data, only every $f=20$ -th step of each trajectory is retained.

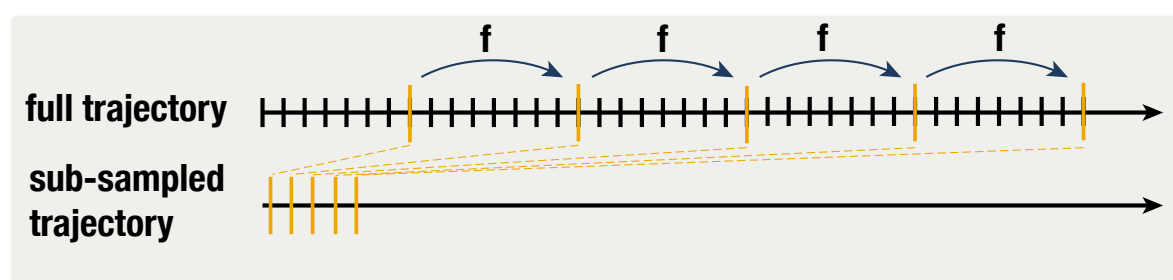


Figure 3. Temporal subsampling of expert demonstrations. Every f -th transition is retained from each expert trajectory ($f=8$ is shown for illustration).

IQ-Learn

Inverse soft-Q learning (IQ-Learn) [1] learns to imitate without the true reward and adversarial training. A single soft Q-function Q_θ encodes a reward by inverting the soft Bellman equation:

$$r_Q(s, a) = Q(s, a) - \gamma \mathbb{E}_{s'}[V_Q(s')]. \quad (1)$$

and the policy is extracted with a Soft Actor-Critic (SAC, [2])-style actor update:

$$\max_{\phi} \mathbb{E}_{s,a \sim \pi_{\phi}} [Q_{\theta}(s, a) - \log \pi_{\phi}(a | s)]. \quad (2)$$

This makes IQ-Learn dynamics-aware and stable to train, at the cost of requiring off-policy environment interactions.

BC

Unlike IQ-Learn, Behavioral Cloning (BC) does not reason about dynamics or future states. It reduces imitation to supervised learning: directly mapping states to actions by maximizing the likelihood of expert actions:

$$\max_{\pi} \mathbb{E}_{(s,a) \sim \pi_E} [\log \pi(a | s)]. \quad (3)$$

BC is therefore fast and requires no environment interaction, but suffers from distribution shift as errors compound outside the expert state distribution. It is used as both a baseline and the warm-start for CSIL.

CSIL

Coherent Soft Imitation Learning (CSIL) [7] builds on BC to recover a shaped reward without adversarial training, in three phases.

- **Phase 1 (BC):** π_{bc} is trained on expert demonstrations.
- **Phase 2 (Coherent reward):** By inverting the entropy-regularised policy update, a shaped reward is derived for which π_{bc} is optimal:

$$\tilde{r}(s, a) = \alpha \log \frac{\pi_{bc}(a | s)}{p(a | s)}, \quad (4)$$

where p is a uniform prior over discrete actions. $\tilde{r}$ is recomputed online from the frozen π_{bc} at every SAC update.

- **Phase 3 (SAC fine-tuning):** SAC is run with $\tilde{r}$ as the sole reward signal. The actor is initialized from π_{bc} , making the coherent reward immediately informative. A pessimistic twin Q-critic with automatic entropy tuning is used for stability.

CSIL + SOAR

Imitation Learning with Soft Optimistic Actor cRitic (IL-SOAR) [6] enhances CSIL by replacing the twin critic architecture with an ensemble of L independent Q-networks to drive principled exploration. SAC's entropy regularisation alone leaves unvisited state-action pairs untargeted [6]. The optimistic value overestimates state-action pairs with high uncertainty:

$$Q^{\text{opt}}(s, a) = \bar{Q}(s, a) + \text{clip}(\sigma_Q(s, a), 0, \sigma), \quad (5)$$

where $\bar{Q} = \frac{1}{L} \sum_{\ell} Q_{\ell}$ and σ_Q is the standard deviation across the ensemble. High disagreement increases the value of uncertain regions, pushing the policy to explore them. The coherent reward $\tilde{r}$ from CSIL is kept unchanged.

Experimental Setup

- **Environments:** CartPole-v1 and Acrobot-v1, expert dataset sizes $K \in \{1, 3, 7, 10, 15\}$, with temporal subsampling $f = 20$ (every 20th transition retained)
- **Seeds:** 5 training seeds, evaluation averaged over 5 evaluation seeds $\times$ 20 episodes = 100 episodes per training seed
- **Networks:** 2-layer MLPs, 64 hidden units (all methods)
- **IQ-Learn:** 100k / 200k gradient steps (CartPole / Acrobot)
- **CSIL / CSIL+SOAR:** 1500 SAC episodes, BC pre-training for 600 epochs
- **SOAR:** optimistic ensemble of $L = 4$ critics, std clipping $\sigma = 10$

Results

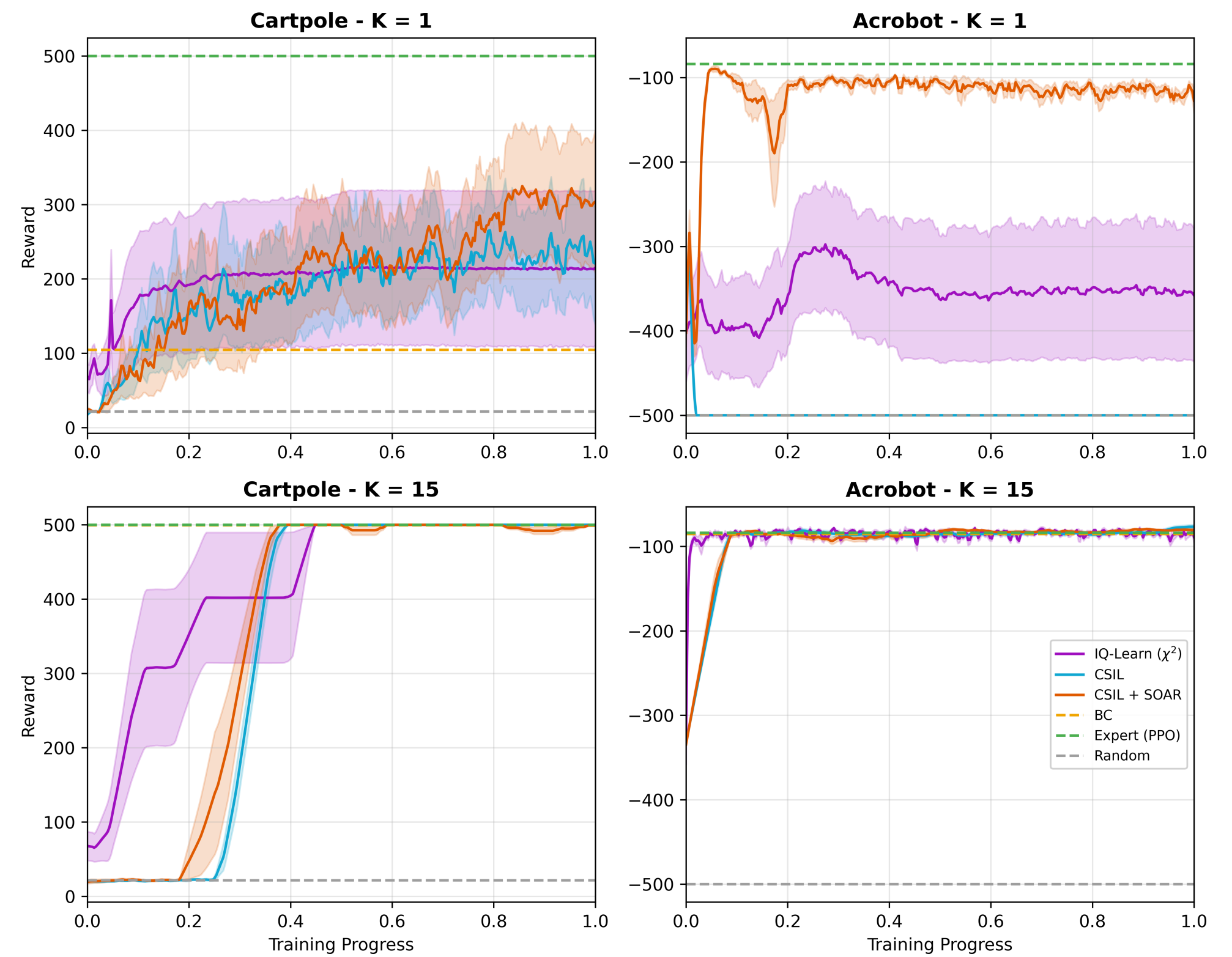


Figure 4. Training curves for IQ-Learn (χ^2), CSIL, and CSIL+SOAR on **CartPole-v1** and **Acrobot-v1**, with $K = 1$ (top) and $K = 15$ (bottom) expert trajectories. Shaded bands show ± 1 standard error of the mean over 5 training seeds. BC, Expert (PPO), and Random baselines are shown as dashed horizontal lines.

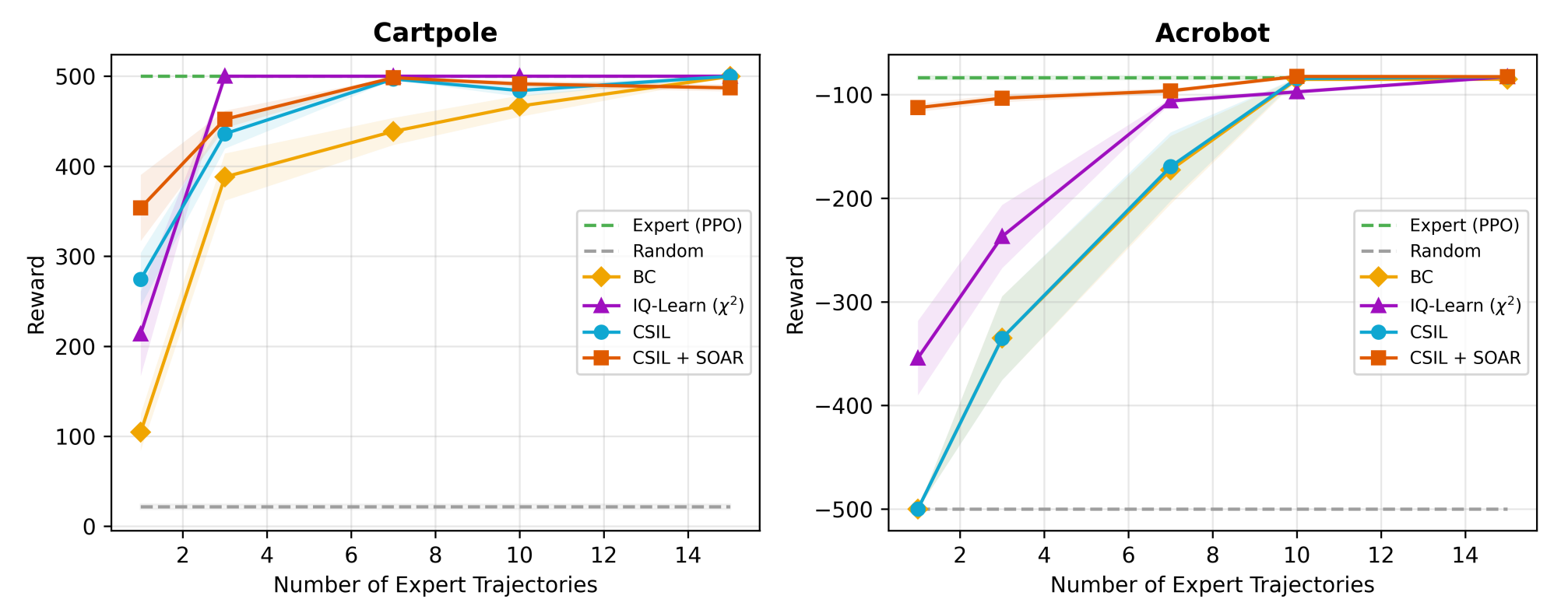


Figure 5. Evaluation performance of BC, IQ-Learn (χ^2), CSIL, and CSIL+SOAR: mean episode return as a function of the number of expert trajectories K , for **CartPole-v1** and **Acrobot-v1**. Shaded bands show ± 1 standard error of the mean over 25 seeds (5 training $\times$ 5 evaluation). Expert (PPO) and Random baselines are shown as dashed horizontal lines.

Discussion

IQ-Learn

- **CartPole-v1:** The expert achieves the maximum return of 500 with near-zero variance, the random baseline is low with high variance, as the stochastic initial pole angle determines how long it accidentally survives. IQ-Learn reaches ≈ 500 for $K \geq 3$, solving the task consistently. At $K = 1$, only ≈ 25 subsampled transitions are available and the agent fails to converge reliably within the training budget.
- **Acrobot-v1:** The random baseline scores exactly -500 with zero variance, it never swings up, exhausting the time limit every episode. The expert shows visible variance since even an optimal policy requires a variable number of steps to time the swing. IQ-Learn solves Acrobot reliably only from $K \geq 7$ and variance decreases with more demonstrations.

CSIL

- **CartPole-v1:** CSIL reaches ≈ 500 for $K \geq 7$, matching the expert. At $K = 1$, a single trajectory produces a weak BC policy and thus a poorly shaped reward $\tilde{r}$, causing unreliable convergence. Although, CSIL manages to outperform plain BC systematically, meaning having access to environment dynamics reduced covariate shifts obtained from BC.
- **Acrobot-v1:** CSIL shows no further improvement compared to plain BC. This result can be explained by the task context paired with the pessimistic twin-critics it leverages: to attain the expert performance, CSIL has to significantly diverge from the BC-derived expert policy, which is itself discouraged by the pessimistic critics.

CSIL+SOAR

- **CartPole-v1:** By enhancing CSIL with SOAR, the performances increased compared to vanilla CSIL, while still being slower to converge than IQ-Learn ($K \geq 7$ vs $K \geq 3$). This is expected due to the optimistic, exploration-driven behavior the ensemble of critics offers, and its heavy reliance on good trajectories during the BC phase. The small monotonic decrease starting from $K \geq 10$ is most likely due to training seed variance.
- **Acrobot-v1:** Compared to vanilla CSIL, its enhanced version clearly outperforms BC and even IQ-Learn (-120 reward vs -500 and -350 respectively), showing close to expert performance with only 1 trajectory. This shows how the optimistic ensemble of critics can significantly help the agent explore uncertain states while following the expert direction.

Conclusion

Different imitation learning algorithms were compared and their performance were assessed on two classical control environments. IQ-Learn showed strong robustness to sparse expert data, while CSIL+SOAR demonstrated that an optimistic critic ensemble can effectively compensate for a conservative BC policy, achieving near-expert performance even at low K .

References

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